Xinyu Su

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Education

Georgia Institute of Technology Atlanta, GA

B.S. in Computer Science (AI & Media), Minor in Industrial Design — 3.64/4.00 GPA *Expected Spring 2028*

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Artificial Intelligence, Computer Graphics, Machine Learning, Design & Analysis of Algorithms, Visual Design Thinking, Design Methods

Projects

SpeechWrite | *Python, FastAPI, pandas, OpenAI API* February 2026

- Built a tool measuring bias in automatic speech recognition across 5 English accent groups, benchmarking OpenAI's speech-to-text model against Mozilla Common Voice ground-truth transcripts
- Engineered the evaluation pipeline computing word error rate per clip, averaged over 5 passes to control for model nondeterminism, with incremental processing to skip already-scored audio
- Mapped transcription errors to CMU dictionary phonemes and quantified substitutions and deletions, revealing 2-5x higher error rates for Indian and South African English

Adversarial Game AI Agent | *Python, NumPy* April 2026

- Built a game-playing agent for a two-player stochastic board game using negamax with alpha-beta pruning, iterative deepening to depth 8, and move ordering under a 4-minute total time budget
- Implemented a hidden Markov model tracking a moving hidden target across 64 cells, applying Bayesian updates from noisy sound and distance readings to a stationary-distribution prior

Empluzz Recruitment | *Python, Django, SQL, HTML/CSS* August 2025 – December 2025

- Served as Product Owner across 3 sprints, scoping 20+ user stories and prioritizing backlog with a client-facing TA
- Built role-based access control on a custom user model with a reusable decorator gating protected views, redirecting unauthorized users and conditionally rendering navigation per role
- Implemented the job posting module: full CRUD over a schema covering salary range, remote status, visa sponsorship, and skill tags, with multi-field search over the same fields

Experience

Undergraduate Researcher August 2026 – Present

WELab Vertically Integrated Project (VIP) *Atlanta, GA*

- Building the work order module for a React, FastAPI, and PostgreSQL facility management app for 100+ parishes, covering request intake, technician claiming, and a 6-state status workflow from unassigned to manager-closed
- Designing a work order schema linking each request to a building asset so every property keeps a maintenance history, with priority-based due dates and required completion notes for low-tech staff

Technical Lead Instructor May 2026 – August 2026

iCode STEM School *Augusta, GA*

- Deployed Veyon classroom management across the Windows lab, resolving networking and permissions issues
- Automated generation of internal deployment and configuration documentation using Python and ReportLab
- Documented instructor onboarding; taught Python, Unity, Unreal Engine 5, and Roblox Studio to 100+ students

Full Stack Developer August 2024 – May 2025

Georgia Institute of Technology | Web Development Club *Atlanta, GA*

- Built LoFiLoop's realtime chat backend with Socket.io and Express, broadcasting messages to all connected clients over a persistent WebSocket channel and storing chat history and a shared song queue in MongoDB
- Designed a 3-command slash system routing user input to backend handlers, integrating the OpenAI API with JSON-mode responses to convert natural-language song requests into structured title and artist data
- Built CollabJournal's landing and authentication pages, porting static markup into React components with react-router navigation, and resolved global CSS collisions across component stylesheets

Technical Skills

Languages: Java, Python, JavaScript, Typescript, HTML/CSS
Frameworks: Django, FastAPI, React, Node.js, Next.js, REST APIs, pandas, NumPy
Tools: Git, Figma, Adobe Photoshop, Vercel, MongoDB, PostgreSQL